

Accessible Slides in LaTeX

https://github.com/rhstanton/accessible_LaTeX

Version 1.5

Richard Stanton

richard.stanton@berkeley.edu

UC Berkeley

March 20, 2026

L^AT_EX and accessibility: new requirements

- From **April 2026**, course materials must meet WCAG 2.1 Level AA accessibility standards
 - Applies even to password-protected course sites (e.g., **bCourses**).
- Many of us use L^AT_EX to create teaching materials, both slides and documents.
- **But standard L^AT_EX (especially Beamer) does not automatically generate accessible PDFs.**
- **Good news:** The L^AT_EX Project team has developed solutions.

What is this project?

- A practical template set for accessible $\LaTeX$ using the [\$\LaTeX\$ Tagging Project](#)
 - **Two templates:** Articles (**article** class) and slides (**ltx-talk**)
 - **Shows:** What is shared and what differs
 - **Contains:** Math, text, figures, and tables
 - **Validation status:** Slides score 100% with [Ally](#) and pass common validators ([veraPDF](#), [PAC](#), [Acrobat](#)) in typical workflows (tested March 17, 2026).
- Also includes some (optional) additional slide features
 - Readable [outline slide](#)
 - Automatic [section separator slides](#)
 - Multi-page [bibliography support](#)
- Repo: https://github.com/rhstanton/accessible_LaTeX

Converting to accessible $\text{\LaTeX}$: the big picture

- **Your existing $\text{\LaTeX}$ skills transfer:** mainly adding a few consistent accessibility features.
- **Most changes are local:** metadata once; then per-figure and per-table tweaks.
- **Applies to both slides & articles:**
 - Document metadata + PDF tagging
 - Alt text on images
 - Table header row tagging
 - Sufficient color contrast
 - Compile with Lua $\text{\LaTeX}$
- **Key fork:**
 - Articles: **keep your existing class** and add tagging features
 - Slides: switch to `ltx-talk` (frame syntax similar to Beamer), then recreate styling

Outline

1. Setup
2. Core requirements
3. Examples
4. Slides versus articles
5. Validation and known issues
6. Quick start
7. (Optional) additional utilities
8. References

1. Setup

Version requirements: what actually matters

- **Engine:** must compile with [Lua \$\LaTeX\$](#) (not pdf $\LaTeX$ or Xe $\LaTeX$).
- **LaTeX kernel/format date:** must be [2025-11-01 or later](#).
- **Important:** a “TeX Live year” alone is not a reliable guarantee.
 - You need a sufficiently new $\LaTeX$ kernel *on your install*, or use Overleaf Labs.
- **Quick check (in any document preamble):**

```
\show\fmtversion
```

Overleaf (works when configured correctly)

- `ltx-talk` requires a very recent $\text{\LaTeX}$ kernel (format date 2025-11-01+).
- On Overleaf, this typically means using the Labs program (not standard Overleaf).
- **Overleaf setup (2 steps):**
 1. Join Overleaf Labs:
 - <https://www.overleaf.com/labs/participate>
 - Opt in and enable Rolling TeX Live releases
 2. Configure the project:
 - TeX Live version: Rolling TeXLive (labs) (bottom of list)
 - Compiler: Lua $\text{\LaTeX}$
- Overleaf guide: <https://docs.overleaf.com/writing-and-editing/creating-accessible-pdfs>

2. Core requirements

Core requirement 1: document metadata

- Required for both slides and articles
- Every accessible document **must** begin with \DocumentMetadata
- Goes **before** \documentclass

```
\DocumentMetadata{
  % Request both PDF/A archival conformance and PDF/UA accessibility tagging
  pdfstandard=a-2u,      % PDF/A-2u format
  pdfstandard=ua-1,      % Add PDF/UA-1 conformance target
  pdfversion=1.7,        % Explicit PDF version
  lang=en-US,            % Language for screen readers
  tagging=on,            % Enable PDF tagging
  tagging-setup={
    math/alt/use,         % Keep math accessibility enabled
    math/mathml/AF=false, % Do not embed MathML as PDF Associated Files
    math/tex/AF=false,    % Do not embed TeX source as Associated Files
    math/mathml/sources=  % Clear MathML source attachment list
  }
}
```

- **Why PDF 1.7 /UA-1?** Current checkers (PAC, Acrobat, Ally, veraPDF) are most reliable for PDF/UA-1 and may incorrectly flag valid PDF 2.0 / UA-2 features
 - See [PDF Days 2025 \(LaTeX Tagged PDF talk\)](#)

Core requirements 2–4: images, tables, and color contrast

- **Required for both slides and articles.**
- **Images:** every `\includegraphics` needs `alt={...}`.
- **Tables:** mark header rows using `\tagpdfsetup{table/header-rows={...}}`.
- **Colors:** WCAG 2.1 AA requires 4.5:1 contrast ratio for normal text.
- **Accessible color defaults:**

```
\colorlet{AccessibleRed}{red!80!black}  
\colorlet{AccessibleGreen}{green!40!black}  
% Standard blue is fine
```

Core requirement 5: compile with Lua $\text{\LaTeX}$

- **Required for both slides and articles.**
- Compile with **Lua $\text{\LaTeX}$** (not pdf $\text{\LaTeX}$ or Xe $\text{\LaTeX}$).
- **Why Lua $\text{\LaTeX}$?**
 1. Automatic math accessibility support (no manual MathML files in normal workflows)
 2. Full tagging support
 3. Good Unicode/OpenType font support
- **How to switch:**
 - Command line: `lualatex filename.tex`
 - Editors: select LuaLaTeX as the compiler

3. Examples

Slides content: mostly the same as Beamer

- Slides live in `frame` environments (similar to Beamer).
- So your content often transfers with minimal changes.

- **Typical frame skeleton:**

```
\begin{frame}{Title}  
  \begin{itemize}  
    \item First point  
  \end{itemize}  
\end{frame}
```

- **Note (optional):** template fonts keep digits/percent/dollar signs consistent in math/text.
 - **Text:** \$1234567890%.
 - **Math:** \$1234567890%.

Figures: alt text

- Figures are included as usual (e.g., `\includegraphics`), but **add alt text**.

- Good (minimal):** `\includegraphics[alt={A capybara}]{capybara.jpg}`

- Better (more informative):**

```
\includegraphics[alt={A capybara's head facing left}]{capybara.jpg}
```

- Decorative images: use empty alt text: `alt={}`.



Tables: header rows

- Tables use tabular as usual, but you must specify **header rows**.
- Use {1} for one header row, {1,2} for two header rows, etc.
- **Note:** PAC may still warn about header associations in some cases; see [Validator-specific notes](#).

```
\tagpdfsetup{table/header-rows={1,2,3}}  
\begin{tabular}{cccrcccr}  
\toprule  
← 3 header rows  
\midrule  
← data rows  
\bottomrule  
\end{tabular}
```

Payment date	Caplet expiry date	DF_{pay}	Forward rate	Days to expiry	Days in accrual period	T_{expiry}	Δ	Caplet
2004/12/01	—	0.99550	0.01790	0	91	0.00000	0.25278	—
2005/03/01	2004/11/29	0.99008	0.02188	89	90	0.24384	0.25000	1,178.77
2005/06/01	2005/02/25	0.98401	0.02413	177	92	0.48493	0.25556	4,844.73
2005/09/01	2005/05/27	0.97733	0.02675	268	92	0.73425	0.25556	10,016.71

4. Slides versus articles

Articles vs. slides

- **Articles:** keep your existing class (e.g., `article`) and add the checklist items.
- **Slides:** switch to `ltx-talk`.
 - Your `\begin{frame}` content often stays the same.
 - Remove Beamer themes/templates; recreate styling with standard $\text{\LaTeX}$ packages (as in this template).
- **If you're converting Beamer slides:**

<code>\documentclass{ltx-talk}</code> (instead of beamer)

Getting started (new vs. converting): keep it simple

- **New documents:**
 - copy `accessible_slides.tex` or `accessible_article.tex`
 - replace content; keep the metadata/preamble structure
- **Converting existing documents:**
 - add metadata + Lua^AT_EX + alt text + table header rows
 - for slides, switch to `ltx-talk` and redo styling once
- Repo: https://github.com/rhstanton/accessible_LaTeX

5. Validation and known issues

Common pitfalls

- **Confusing PDF tagging with PDF bookmarks**
 - tagging=on creates the structure tree for **screen readers** (required).
 - PDF bookmarks (navigation pane) are separate and optional.
- **Forgetting alt text**
 - Every `\includegraphics` needs `alt={...}` (or `alt={}` if decorative).
- **Not tagging table header rows**
 - Add `\tagpdfsetup{table/header-rows={...}}` before each table.
- **Low contrast colors**
 - Avoid light colors (e.g., plain yellow/cyan).
 - Use darker variants for red/green (defined in this template).
- **Wrong engine or old kernel**
 - Use Lua $\text{\LaTeX}$ and ensure format date 2025-11-01 or later.

Validation workflow: use multiple checkers

- No single checker catches everything.
- Accessibility usability checks and strict PDF conformance checks are not identical.
- Practical workflow: run 2+ tools and compare findings.
- **Common tools:**
 - Ally (bCourses)
 - veraPDF (PDF/A conformance): <https://verapdf.org/>
 - PAC (PDF accessibility): <https://pac.pdf-accessibility.org/>
 - Adobe Acrobat (useful but has known quirks)

Optional cleanup for strict validators: `strip_af.py`

- The metadata settings alone can be enough for a 100% score in Ally.
- Some strict tools (e.g., [veraPDF](#)) may still complain about /AF or embedded-file structures.
- To remove these, run:

```
python strip_af.py build/accessible_slides.pdf
```

- Prerequisite: install the Python package [pikepdf](#) first: `python -m pip install pikepdf`.
- Then validate again with veraPDF.

Validator-specific notes (common false positives)

- **Acrobat: “Lbl and LBody – Failed” (validator bug)**
 - Often appears around figure captions, section numbers, table captions.
 - This is an Acrobat bug in common setups; typically safe to ignore if source tagging is correct.
 - Reference: <https://tex.stackexchange.com/a/709655/2388>
- **PAC table warning (known behavior)**
 - PAC can report: Table header cell has no associated subcells even when table/header-rows is correct.
 - Documented by the Tagging Project:
<https://github.com/latex3/tagging-project/issues/1056>
- **Bottom line:** treat one-tool warnings as investigation prompts, not definitive errors.

6. Quick start

Quick start: accessible $\text{\LaTeX}$ on one slide

- **To make an accessible $\text{\LaTeX}$ PDF:**
 1. Add `\DocumentMetadata{...}` (before `\documentclass`)
 2. Compile with $\text{\Lua\LaTeX}$
 3. Every figure: `\includegraphics[alt={...}]{...}`
 4. Every table: `\tagpdfsetup{table/header-rows={...}}`
 5. Use high-contrast colors (4.5:1) and validate with 2+ tools
- Template repo: https://github.com/rhstanton/accessible_LaTeX
- Questions: richard.stanton@berkeley.edu

7. (Optional) additional utilities

Optional presentation utilities

- This template includes **optional utilities** in `slide_utils.sty`
 - **Not required for accessibility**
 - Useful for common presentation tasks
 - Easy to customize or remove
- **Two optional features:**
 1. Automatic section separator slides
 2. Multi-page bibliography support
- See `slide_utils.sty` for implementation details and customization options.
- **Note:** There's additional code in the main preamble to get readable table of contents.

Optional feature 1: section separator slides

- **What it does:** Automatically inserts a separator slide after each `\section` command
 - Displays section number and title in large text
 - Centered on the slide for visual emphasis
 - Uses accessible colors (AccessibleRed for sections, blue for subsections)
- **Example:** After `\section{Core requirements}`, automatically creates a slide saying

3. Core requirements

- **To disable:** Comment out or remove the redefinition code in `slide_utils.sty` Section 1
- **To customize:** Edit colors, fonts, or spacing in the frame template

Optional feature 2: multi-page bibliography

- **Problem:** `\bibliography` inside frame environment puts all references on one slide.
 - For large bibliographies, this can be unreadable.
 - This was easy with Beamer, using the `allowframebreaks` option.
 - But `ltx-talk` does not have a similar option.
- **Solution:** Write code to split bibliography automatically across multiple slides
 - Works with standard BibTeX workflow
 - Uses Lua script ([rsbibsplitsplit.lua](#)) to split the .bbl file
 - (Lua^ATeX provides access to the [Lua](#) programming language)
- **Usage:** As usual (see end of this .tex file for complete example)
 - `\cite{key1,key2,...}` to specify references
 - `\bibliography{bibfile}` to specify bib file
 - **Note:** Do not put `\bibliography` command inside frame environment
 - (Set `\RS@bibmax` to customize max per slide; default=8).
- See `slide_utils.sty` Section 2 for detailed workflow explanation.

8. References

References

- Akerlof, G. A., 1970, The market for “lemons”: Quality uncertainty and the market mechanism, *Quarterly Journal of Economics* 84, 488–500.
- Arrow, K. J., 1963, Uncertainty and the welfare economics of medical care, *American Economic Review* 53, 941–973.
- Barro, R. J., 1991, Economic growth in a cross section of countries, *Quarterly Journal of Economics* 106, 407–443.
- Black, F., and M. Scholes, 1973, The pricing of options and corporate liabilities, *Journal of Political Economy* 81, 637–654.
- Coase, R. H., 1960, The problem of social cost, *Journal of Law and Economics* 3, 1–44.
- Fama, E. F., 1970, Efficient capital markets: A review of theory and empirical work, *Journal of Finance* 25, 383–417.
- Fama, E. F., and K. R. French, 1993, Common risk factors in the returns on stocks and bonds, *Journal of Financial Economics* 33, 3–56.
- Jensen, M. C., and W. H. Meckling, 1976, Theory of the firm: Managerial behavior, agency costs and ownership structure, *Journal of Financial Economics* 3, 305–360.

References

- Kahneman, D., and A. Tversky, 1979, Prospect theory: An analysis of decision under risk, *Econometrica* 47, 263–292.
- Krugman, P. R., 1991, Increasing returns and economic geography, *Journal of Political Economy* 99, 483–499.
- Lucas, R. E., Jr., 1976, Econometric policy evaluation: A critique, *Carnegie-Rochester Conference Series on Public Policy* 1, 19–46.
- Merton, R. C., 1974, On the pricing of corporate debt: The risk structure of interest rates, *Journal of Finance* 29, 449–470.
- Modigliani, F., and M. H. Miller, 1958, The cost of capital, corporation finance and the theory of investment, *American Economic Review* 48, 261–297.
- Rothschild, M., and J. E. Stiglitz, 1976, Equilibrium in competitive insurance markets: An essay on the economics of imperfect information, *Quarterly Journal of Economics* 90, 629–649.
- Solow, R. M., 1956, A contribution to the theory of economic growth, *Quarterly Journal of Economics* 70, 65–94.